

Mohamed Ebrahim Mahfouz

Data Scientist

+201013291430 | mohamede121204@gmail.com | Asyut, Egypt | [linkedin.com/in/mohamed-ibrahim-1m](https://www.linkedin.com/in/mohamed-ibrahim-1m)

SUMMARY

Computer Science student looking to launch a career in data analysis. Built end-to-end ML pipelines using Python, pandas, and scikit-learn — including a heart disease prediction model with 90%+ accuracy. Bringing strong analytical thinking and a genuine eagerness to contribute from day one.

EDUCATION

Asyut National University, Asyut, Egypt

2023 — Present

GPA: 3.4 / 4.0

PROJECTS

EduScript — Custom Programming Language Compiler & Transpiler

- Designed and implemented a custom programming language compiler in Python, including a lexer, recursive-descent parser, and semantic analyzer with scoped type checking
- Built a source-to-source transpiler converting the AST into syntactically valid Python using the Visitor design pattern, supporting 15+ language constructs validated against 20+ test programs
- Developed an interactive Streamlit IDE with live Tokens, AST, Semantic Analysis, and Generated Python views

Heart Disease Prediction — Machine Learning Pipeline

- Built an end-to-end ML pipeline in Python (pandas, scikit-learn) on a dataset of ~300 patient records, improving accuracy from ~82% (baseline Logistic Regression) to 90%+ after tuning
- Trained and compared 5 classification models (Logistic Regression, KNN, SVM, Naive Bayes, Random Forest) with hyperparameter tuning via GridSearchCV
- Performed EDA with correlation heatmaps, feature distributions, and scatter plots using matplotlib and seaborn

Tic-Tac-Toe AI Game — Minimax Algorithm

- Developed a desktop Tic-Tac-Toe game in Python with Tkinter, featuring an unbeatable AI using the Minimax algorithm, achieving a 100% unbeatable win/draw rate across 50+ test games on Hard difficulty
- Implemented heuristic-based AI logic for adjustable difficulty (Easy/Medium/Hard) with full win/draw detection

SKILLS

Technical Skills: C#, Power BI, C++, Python, Object-oriented programming, Database, SQL, Data Structure, HTML & CSS, Git & GitHub

Soft Skills: Self-learning, Problem-solving, Dedication to continuous learning, Communication, Collaboration, Commitment and Hard Work, Adaptability

Languages: Arabic (Native), English (B1)

EXPERIENCE

Data Scientist — DEPI

Jul 2026 — Present

Data Analysis Trainee — Bootcamp — Asyut National University (Completed)

- Successfully completed an 80+ hour Data Analysis Bootcamp program at the university, gaining practical experience in data analysis and data-driven problem solving
- Applied data analysis techniques to a graduation project, working through the process from data preparation and analysis to presenting the final results
- Developed practical skills in Python, pandas, data visualization, and analytical thinking

Data Scientist Trainee — Digital Egypt Pioneers Initiative (DEPI)

2026 — Present

- Currently participating in the Data Science track, developing practical skills in data analysis, machine learning, and data science workflows
- Applying Python and machine learning techniques through hands-on training and practical projects
- Expected to complete the program and graduate as a Data Scientist

Data Science & AI Trainee — GCI — Data Science & AI Track

2026 — Present

- Selected to participate in the Data Science & AI track, gaining hands-on exposure to data science and artificial intelligence concepts
- Developing practical skills through technical training and project-based learning
- Expected to complete the program in 2026